

ACTL3182 Wk1

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Introduction of me

My name is Tadhg (tie-gh, T, TXG).

According to my bio (ACTL3182 Moodle > Course Information > Teaching Staff and Contacts), Tadhg is in his fourth year of Actuarial Studies and Mathematics and has tutored a number of courses, with ACTL3182 being his sixth. In his spare time, he enjoys answering questions on the Ed forum and eating good food.

Hard stocks and bonds course

A utility function $U(\cdot)$ is used to represent how much an individual values something, typically wealth, with larger values representing more value.

Typically, we have

- $U'(\cdot) > 0$ - the individual always prefers more wealth;
- $U''(\cdot) < 0$ - the individual is risk averse.

Expected Utility Framework

An individual with utility function $U(\cdot)$ prefers risk X_1 over X_2 if and only if

$$\mathbb{E}[U(X_1)] > \mathbb{E}[U(X_2)].$$

The individual is indifferent between the risks if there is equality.

Certainty Equivalent

Given a random risk X , the certainty equivalent $c \in \mathbb{R}$ is the level of wealth where the individual is indifferent to taking the risk or the guaranteed amount of wealth, that is,

$$U(c) = \mathbb{E}[U(X)].$$

Absolute and Relative Risk Aversion

Let

$$A(w) = -\frac{U''(w)}{U'(w)}, \quad R(w) = w A(w)$$

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$A(w)$ shows how risk aversion changes as wealth increases

→ if $A(w)$ is increasing in wealth, then the individual becomes more risk averse as their wealth increases and the nominal *amount* invested in risky assets is decreasing.

$R(w)$ is how much is invested in risky assets as wealth increases

→ if $R(w)$ is increasing, then risk aversion is increasing and thus the *proportion* of wealth invested in risky assets is decreasing.

Examples of Risk Measures

Let X be a random return with mean μ and pdf f_X .

Semi-variance: variance below the mean,

$$\text{semi-Var}(X) = \int_{-\infty}^{\mu} (x - \mu)^2 f_X(x) dx.$$

Shortfall-Probability: for some threshold ℓ ,

$$p_{\ell}(X) = \mathbb{P}(X < \ell).$$

Expected-Shortfall: Shortfall below some threshold ℓ ,

$$\text{ES}_{\ell}(X) = \mathbb{E}[(\ell - X)^+] = \int_{-\infty}^{\ell} (\ell - x) f_X(x) dx$$

VaR

Let X be a risk and $\alpha \in (0, 1)$. Then,

$$\text{VaR}_{1-\alpha}(X) = \inf\{x : \mathbb{P}(X > x) \leq \alpha\}.$$

Intuitively, the value of $\text{VaR}_{1-\alpha}(X)$ is such that we are $(1 - \alpha)\%$ sure that our loss won't exceed this value.